

# Problem 2b: Step-by-Step Calculation (Getting $\lambda_1$ )

ECO 310 - Detailed Walkthrough

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## 1 The Setup

**The Lagrangian (Convention 2):**

$$\mathcal{L} = -p_1x_1 - p_2x_2 + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*) \quad (1)$$

**The guess we're testing:**

- Utility constraint is binding:  $x_1 + x_2 = u^*$
- Nonnegativity on good 1 is binding:  $x_1 = 0$
- Nonnegativity on good 2 is non-binding:  $x_2 > 0$

**What this means:**

- Since  $x_1 = 0$ , we're buying NO good 1
- Since  $x_2 > 0$ , we're buying SOME good 2
- Combined with  $x_1 + x_2 = u^*$ , we get  $x_2 = u^*$  (buy only good 2)

This guess makes sense when **good 2 is cheaper**:  $p_2 < p_1$

## 2 The General Optimality Conditions

From taking derivatives of the Lagrangian:

**FOC with respect to  $x_1$ :**

$$\frac{\partial \mathcal{L}}{\partial x_1} = -p_1 + \lambda_1 + \lambda_3 \leq 0, \quad x_1 \geq 0, \quad x_1(-p_1 + \lambda_1 + \lambda_3) = 0 \quad (2)$$

**FOC with respect to  $x_2$ :**

$$\frac{\partial \mathcal{L}}{\partial x_2} = -p_2 + \lambda_2 + \lambda_3 \leq 0, \quad x_2 \geq 0, \quad x_2(-p_2 + \lambda_2 + \lambda_3) = 0 \quad (3)$$

**FOC with respect to  $\lambda_3$ :**

$$\frac{\partial \mathcal{L}}{\partial \lambda_3} = x_1 + x_2 - u^* \geq 0, \quad \lambda_3 \geq 0, \quad \lambda_3(x_1 + x_2 - u^*) = 0 \quad (4)$$

### 3 Step 1: Use Complementary Slackness for $x_2$

Since  $x_2 > 0$  (strictly positive), the complementary slackness condition says:

$$x_2 \cdot (-p_2 + \lambda_2 + \lambda_3) = 0 \quad (5)$$

Because  $x_2 > 0$ , we need:

$$-p_2 + \lambda_2 + \lambda_3 = 0 \quad (6)$$

The term in parentheses MUST equal zero (not just  $\leq 0$ ).

### 4 Step 2: Use Complementary Slackness for $\lambda_2$

We also have complementary slackness for the constraint  $x_2 \geq 0$ :

$$\lambda_2 \cdot x_2 = 0 \quad (7)$$

Since  $x_2 > 0$ , we must have:

$$\lambda_2 = 0 \quad (8)$$

### 5 Step 3: Solve for $\lambda_3$

Substitute  $\lambda_2 = 0$  into equation (6):

$$-p_2 + 0 + \lambda_3 = 0 \quad (9)$$

$$\lambda_3 = p_2 \quad (10)$$

#### Key Result

$$\lambda_3 = p_2$$

This is the marginal cost of utility. Since we're buying only good 2, each extra unit of utility costs  $p_2$ .

### 6 Step 4: Apply the $x_1$ Condition

Since  $x_1 = 0$  (binding constraint), we have the inequality:

$$-p_1 + \lambda_1 + \lambda_3 \leq 0 \quad (11)$$

Substitute  $\lambda_3 = p_2$ :

$$-p_1 + \lambda_1 + p_2 \leq 0 \quad (12)$$

$$\lambda_1 + p_2 \leq p_1 \quad (13)$$

$$\lambda_1 \leq p_1 - p_2 \quad (14)$$

## 7 Step 5: Determine $\lambda_1$

At the optimal solution, this inequality typically **binds** (holds with equality):

$$\lambda_1 = p_1 - p_2 \quad (15)$$

Important!

**The answer is:**

$$\lambda_1 = p_1 - p_2$$

**NOT**  $\lambda_1 = p_2 - p_1$  (I had an error in the previous guide!)

## 8 Step 6: Check Validity

For this solution to be valid, we need  $\lambda_1 \geq 0$ :

$$\lambda_1 \geq 0 \quad (16)$$

$$p_1 - p_2 \geq 0 \quad (17)$$

$$p_1 \geq p_2 \quad (18)$$

**Wait, something seems wrong!**

If we're buying only good 2 ( $x_1 = 0$ ,  $x_2 = u^*$ ), this should happen when good 2 is *cheaper*, meaning  $p_2 < p_1$ .

But our condition says  $p_1 \geq p_2$ , which is the same thing!

So actually, the solution IS valid when  $p_1 > p_2$  (good 1 is more expensive than good 2).

## 9 Concrete Example

**Setup:**

- Good 1 (coffee) costs \$5:  $p_1 = 5$
- Good 2 (tea) costs \$3:  $p_2 = 3$
- Target utility:  $u^* = 10$

**Guess:** Buy only good 2 (tea is cheaper)

- $x_1 = 0$  (no coffee)
- $x_2 = 10$  (10 teas)

**Calculate shadow prices:**

From Step 3:

$$\lambda_3 = p_2 = 3 \quad (19)$$

From Step 5:

$$\lambda_1 = p_1 - p_2 = 5 - 3 = 2 \quad (20)$$

Also:

$$\lambda_2 = 0 \quad (21)$$

Check validity:

$$\lambda_1 = 2 > 0 \quad \checkmark \quad (22)$$

**Economic interpretation:**

- $\lambda_1 = 2$ : If you could buy "negative coffee" (get money back for not buying coffee), each unit would save you \$2

Actually, that doesn't quite make sense. Let me think about this differently...

- $\lambda_1 = 2$  measures how much your cost would *increase* if you were forced to buy 1 unit of good 1
- Since coffee costs \$5 but tea costs \$3, replacing 1 tea with 1 coffee increases cost by  $5 - 3 = 2$  dollars
- $\lambda_3 = 3$ : Marginal cost of utility is \$3 (the price of tea, which you're buying)

## 10 Summary: Step-by-Step Process

### How to Solve Problem 2b

**Given the guess:**  $x_1 = 0, x_2 > 0, x_1 + x_2 = u^*$

**Step 1:** Use complementary slackness

- Since  $x_2 > 0$ : the  $x_2$  FOC holds with equality
- Since  $x_2 > 0$ : we have  $\lambda_2 = 0$

**Step 2:** Solve for  $\lambda_3$

- From  $x_2$  FOC:  $-p_2 + \lambda_2 + \lambda_3 = 0$
- Substitute  $\lambda_2 = 0$ :  $\lambda_3 = p_2$

**Step 3:** Solve for  $\lambda_1$

- From  $x_1$  FOC:  $-p_1 + \lambda_1 + \lambda_3 \leq 0$
- Substitute  $\lambda_3 = p_2$ :  $\lambda_1 \leq p_1 - p_2$
- At optimum:  $\lambda_1 = p_1 - p_2$

**Step 4:** Find  $x_1$  and  $x_2$

- From guess:  $x_1 = 0$
- From utility constraint:  $x_2 = u^*$

**Step 5:** Check validity

- Need  $\lambda_1 \geq 0$ : requires  $p_1 \geq p_2$
- This makes sense: we buy only good 2 when it's cheaper

## 11 Why Does $-p_1 + \lambda_1 + \lambda_3 \leq 0$ Become $\lambda_1 = p_1 - p_2$ ?

Let me break down this algebra very carefully:

**Starting equation:**

$$-p_1 + \lambda_1 + \lambda_3 \leq 0 \quad (23)$$

**Add  $p_1$  to both sides:**

$$\lambda_1 + \lambda_3 \leq p_1 \quad (24)$$

**Subtract  $\lambda_3$  from both sides:**

$$\lambda_1 \leq p_1 - \lambda_3 \quad (25)$$

**Substitute  $\lambda_3 = p_2$ :**

$$\lambda_1 \leq p_1 - p_2 \quad (26)$$

**At the optimum, assume this binds (equality):**

$$\lambda_1 = p_1 - p_2 \quad (27)$$

### Final Answer

Given the guess that  $x_1 = 0$ ,  $x_2 = u^*$ :

$$x_1 = 0 \quad (28)$$

$$x_2 = u^* \quad (29)$$

$$\lambda_1 = p_1 - p_2 \quad (30)$$

$$\lambda_2 = 0 \quad (31)$$

$$\lambda_3 = p_2 \quad (32)$$

Valid when  $p_1 \geq p_2$  (good 1 is at least as expensive as good 2)